SEN0401 midterm v3, Form A - ANSWER KEY (for the grader, not for students)

Form code 30015. Seed 5701. Chapters 1, 2, 3, 4, 5, 6. Questions 1 to 4: 25 points each (100 in all), read from the answer sheet. Question 5: bonus, 10 points, hand-graded from the lines of the paper. Letters, digits and codes are the rows of the answer sheet; a number keeps the stated tolerance; a trace is the printed text. Partial credit for ordering and matching is proportional per correct part.

1a ChoiceFour 4 points chapters 4,5 Bloom 4 [mt2:i14:wif-flag-and-balance]
row 1a **A**

1b ChoiceFour 4 points chapters 2 Bloom 4 [mt2:i05:shop-confirmations]
row 1b **B**

1c ChoiceFourCode 3 points chapters 3 Bloom 4 [mt2:i09:conf-precedence-and-refusal]
row 1c **D**

1d ChoiceFour 4 points chapters 1,5 Bloom 4 [mt2:i01:gox-two-events]
row 1d **C**

1e ChoiceFour 4 points chapters 3,4 Bloom 4 [mt2:i12:rpcauth-and-p2pkh]
row 1e **C**

1f ChoiceFour 3 points chapters 1,2 Bloom 3 [mt2:i02:fee-not-written]
row 1f **B**

1g ChoiceFour 3 points chapters 5 Bloom 4 [mt2:i20:restore-passphrase-labels]
row 1g **A**

2a CodeTrace 4 points chapters 5 Bloom 4 [mt2:i19:trace-gap-limit]
row 2a **15**

2b NumericAnswer 3 points chapters 1,6 Bloom 3 [mt2:i03:toy-chain-supply]
row 2b **value 3900 coins (tolerance 0)**

2c CodeTrace 3 points chapters 3 Bloom 3 [mt2:i11:trace-release-tags]
row 2c **12**

2d NumericAnswer 3 points chapters 4 Bloom 3 [mt2:i13:vanity-factor]
row 2d **value 3364 times (tolerance 0)**

2e CodeTrace 3 points chapters 6 Bloom 4 [mt2:i22:trace-sequence-field]
row 2e **35**

2f NumericAnswer 3 points chapters 1 Bloom 3 [mt2:i04:vwap-third-trade]
row 2f **value 65000 dollars per bitcoin (tolerance 0)**

2g CodeTrace 3 points chapters 6 Bloom 3 [mt2:i21:trace-compactsize]
row 2g **14**

2h NumericAnswer 3 points chapters 5 Bloom 3 [mt2:i17:words-to-entropy]
row 2h **value 192 bits (tolerance 0)**

3a Matching 9 points chapters 5,2,3,6 Bloom 2 [mt2:i18:match-story-lesson]
row 3a1 **D** (2.25 points)
row 3a2 **B** (2.25 points)
row 3a3 **E** (2.25 points)
row 3a4 **F** (2.25 points)

3b Ordering 8 points chapters 2 Bloom 2 [mt2:i06:order-wallet-to-block]
row 3b1 **1** (1.6 points)
row 3b2 **2** (1.6 points)
row 3b3 **4** (1.6 points)
row 3b4 **5** (1.6 points)
row 3b5 **3** (1.6 points)

3c Matching 8 points chapters 3 Bloom 3 [mt2:i10:match-node-options]
row 3c1 **F** (2 points)
row 3c2 **A** (2 points)
row 3c3 **B** (2 points)
row 3c4 **D** (2 points)

4a FillBlankGroup 9 points chapters 6,3 Bloom 2 [mt2:i23:fill-txid-and-weight]
row 4a **45** (5 points; the entry: txid (the transaction identifier, a hash that ignores the witness))
row 4b **37** (4 points; the entry: less (a smaller amount))

4c FillBlankGroup 8 points chapters 2,6 Bloom 2 [mt2:i07:fill-fee-rate-and-set]
row 4c **37** (4 points; the entry: less (a smaller amount))
row 4d **21** (4 points; the entry: unspent (not yet used as an input by a later transaction))

4e FillBlankGroup 8 points chapters 4 Bloom 2 [mt2:i15:fill-hash-commitment]

row 4e **45** (4 points; the entry: commitment (a value that fixes a secret now and lets its owner reveal it later))

row 4f **37** (4 points; the entry: 256-bit hash (the hash function Secure Hash Algorithm 256-bit (SHA256), whose result is 256 bits long))

5a BONUS, 5 points, chapters 2, hand-graded [mt2:b2:mining-check]

Prompt: In two or three sentences, explain why mining is called a search that nobody can shortcut, and why it is still easy for every other node to check a block.

- The search can only be done by trying numbers; there is no shortcut.
- Anyone can check the found number with one hash.
- The answer states why this matters: the work is costly to produce but cheap to verify.

Marking: 3 rubric statements, 5 points in all; how the points split between the statements is for the instructor (open). The answer sheet has no place for this answer.

5b BONUS, 5 points, chapters 1, hand-graded [mt2:b1:no-bank]

Prompt: In two or three sentences, explain who refuses a double spend (the same coin offered to two people) in Bitcoin when no bank stands in the middle, and why that makes the history costly to change.

- Every node checks every rule, so the shared rule refuses the second spend of a coin; no referee decides.
- Proof of work makes each block costly to produce, so rewriting history costs real work.
- The answer connects the two: a rule everybody checks, and a history that is costly to change.

Marking: 3 rubric statements, 5 points in all; how the points split between the statements is for the instructor (open). The answer sheet has no place for this answer.